

Stephanie Vargas Aguilar, Ph.D.
Department of Molecular Biology
UT Southwestern Medical Center
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EDUCATION

- 2013–2018 **Ph.D., Biology and Immunology**
Humboldt University of Berlin, Berlin, Germany
Dissertation: "Investigation of the molecular mechanisms that enable the self-renewal of tissue-resident macrophages"
Mentor: Michael Sieweke, Ph.D.
- 2006–2012 **German Diploma (equivalent to B.A. + M.S.), Molecular Medicine**
Albert-Ludwigs-Universität Freiburg, Freiburg, Germany
Mentor: Rolf Kemler, Ph.D.

RESEARCH EXPERIENCE

- 2025–present **Instructor**
UT Southwestern Medical Center, Dallas, TX
Mentor: Eric Olson, Ph.D.
- 2020–2025 **Postdoctoral Fellow**
UT Southwestern Medical Center, Dallas, TX
Mentor: Eric Olson, Ph.D.
- 2013–2018 **Graduate Student**
Centre d'Immunologie de Marseille-Luminy, Marseille, France
Max Delbrück Center for Molecular Medicine, Berlin, Germany
Mentor: Michael Sieweke, Ph.D.
- 2011–2012 **Undergraduate Research Technician**
Max Planck Institute for Immunobiology and Epigenetics, Freiburg, Germany
Laboratory of Rolf Kemler, Ph.D.
- 2008–2010 **Undergraduate Research Technician**
Albert-Ludwigs-Universität Freiburg, Institute for Molecular Medicine and Cell Research
Laboratory of Thomas Reinheckel, Ph.D.

PUBLICATIONS

1. **Vargas Aguilar S***, Dos Santos M, Tan W, Sanchez-Ortiz E, Chen K, Li H, Xu L, Liu N, and Olson EN*. A developmentally restricted $\gamma\delta$ T cell–IL-17A axis supports mouse heart regeneration, *Cell Reports*, in press. *co-corresponding authors
2. **Vargas Aguilar S***, Cui M*, Tan W, Sanchez-Ortiz E, Bassel-Duby R, Liu N, and Olson EN. The PD-1–PD-L1 pathway maintains an immunosuppressive environment essential for neonatal heart regeneration. *Nature Cardiovascular Research*, March 2024. PMID: 38737787. *co-first authors
3. **Vargas Aguilar S** (as part of the ImmGen Consortium). Network analysis of large-scale ImmGen and Tabula Muris datasets highlights metabolic diversity of tissue mononuclear phagocytes. *Cell Reports*, February 2023. PMID: 36708514
4. Subramanian S, Busch CJ, Molawi K, Geirsdottir L, Maurizio J, **Vargas Aguilar S**, Belahbib H, Gimenez G, Yuda RAA, Burkon M, Favret J, Gholamhosseinian Najjar S, de Laval B, Kandalla PK, Sarrazin S, Alexopoulou L, Sieweke MH. Long-term culture-expanded alveolar macrophages

restore their full epigenetic identity after transfer in vivo. *Nature Immunology*, March 2022. PMID: 35210623

5. **Vargas Aguilar S** (as part of the ImmGen Consortium). ImmGen at 15. *Nature Immunology*, July 2020
6. Imperatore F, Maurizio J, **Vargas Aguilar S**, Busch CJ, Favret J, Kowenz-Leutz E, Cathou W, Gentek R, Perrin P, Leutz A, Berruyer C, Sieweke MH. SIRT1 regulates macrophage self-renewal. *EMBO Journal*, August 2017. PMID: 28701484
7. Matcovitch-Natan O*, Winter DR*, Giladi A, **Vargas Aguilar S**, Spinrad A, Sarrazin S, Ben-Yehuda H, David E, Zelada González F, Perrin P, et al. Microglia development follows a stepwise program to regulate brain homeostasis. *Science*, August 2016. PMID: 27338705. *co-first authors
8. Dennemärker J, Lohmüller T, Müller S, **Vargas Aguilar S**, Tobin DJ, Peters C, Reinheckel T. Impaired turnover of autophagolysosomes in cathepsin L deficiency. *Biological Chemistry*, August 2010. PMID: 20536383

FUNDING

- 2025–present **Career Development Award | PI | \$231,000**
American Heart Association
- 2022–2024 **Postdoctoral Fellowship | \$145,020**
American Heart Association
- 2016–2017 **Doctoral Scholarship | €49,300**
Foundation ARC pour la Recherche sur le Cancer — Aides doctorales en France
- 2013–2016 **Doctoral Scholarship | €79,457**
International PhD Programme, Helmholtz Graduate School

MENTORSHIP EXPERIENCE

- 2023 Yu-Chung “Andy” Pien – Graduate Student, laboratory of Dr. Eric Olson
- 2017 Sethuraman Subramanian – Graduate Student, laboratory of Dr. Michael Sieweke

PRESENTATIONS

Oral Presentations

- 2024 II. Cardioimmunology Congress, Würzburg, Germany
- 2023 NPAW Symposium 2023, UT Southwestern Medical Center, Dallas, USA
- 2022 Charlas Virtuales CISEAL, Center for Research on Health in Latin America, Quito, Ecuador
- 2022 EMBO Workshop on Molecular and Cellular Basis of Regeneration and Tissue Repair, Barcelona, Spain
- 2020 Immunity in Regeneration Kick-off Mini Symposium, Center for Regenerative Therapies, Dresden, Germany
- 2019 33rd Annual Conference of the European Macrophage and Dendritic Cell Society, Marseille, France
- 2015 Scientific Retreat, Centre d'Immunologie Marseille-Luminy, Marseille, France

Poster Presentations

- 2026 BCVS Scientific Sessions, American Heart Association, Boston, USA

2023	BCVS Scientific Sessions, American Heart Association, Boston, USA
2022	NPAW Symposium 2022, UT Southwestern Medical Center, Dallas, USA
2018	EMBO Workshop “piRNAs and PIWI Proteins”, Montpellier, France
2017	Nature Conference on Regeneration, Milan, Italy
2016	NeuroMac, Freiburg, Germany
2016	EMBO Workshop “Multiple functions of piRNAs and PIWI Proteins”, Montpellier, France

PROFESSIONAL SERVICE

2023–2024	Vice President UT Southwestern Postdoctoral Association
2021–present	Chair of the Webinar Committee for “The Americas” International Society for Regenerative Biology

PROFESSIONAL MEMBERSHIPS

American Heart Association
 International Society for Heart Research
 International Society for Regenerative Biology
 Latin American Society for Developmental Biology
 Organization for Women in Science for the Developing World

SCIENTIFIC OUTREACH

2023	My personal and scientific path as a minority in STEM Invited speaker , Hispanic Heritage Month Symposium, UT Southwestern Medical Center, Dallas, TX
2022	El anhelo de volver al corazón infantil Invited speaker (virtual) , Villaviciosa ConCiencia Science Outreach Association, Villaviciosa, Spain

OTHER SCHOLARLY PUBLICATIONS

Freire-Fierro A, Pareja-Mejía D, Diele-Viegas LM, Castillo-Briceño P, Martinez-Bengochea A, Henriquez-Coronel P, Morales JF, Bravo-Cazar AL, Novoa Sepúlveda C, Rodrigues KF, **Vargas Aguilar S**, Garcia-Yanes F, Arias T. Brain drain or brain circulation? A perspective from Latin American scientists. *Frontiers in Research Metrics and Analytics*, July 2026. PMID: 42528971

Dallas, October 2026